Base station physics: summary

Params: base/spec/params.py v1.2.9

Spring (each, A228, SF 1.2 on Shigley static limits)

Rate used / by formula	0.125 / 0.117 N/mm
Initial tension used	7.0 N
Body coils, mass	98, 53 g
Safe force (governed by)	40.9 N (hook torsion)
Safe extension (wire) / spec cap	271 / 265 mm
Surge (fixed-fixed)	24 Hz

Launch (chain spring model)

Stroke, ratio, pitch	241 mm, 1.0:1, 35 deg
Spring extension full / rest	265 / 24 mm
Stored energy (2 springs)	12.1 J
Carriage force at full draw	80 N
Muzzle speed	6.80 m/s
Peak acceleration, stroke time	18 g, 59 ms
Efficiency (dart KE / stored)	53 %
Carriage KE at stop, pad peak	1.98 J, 295 N (19 mm pad)
Spring KE at stop, coil clash	0.82 J, 8.1 m/s

Flight (bead-chain tether + reel model, level ground)

Line path at the shot	taut (path only lengthens)
Range at 0.5 / 0.75 / 1.0 N brake	4.13 / 3.72 / 3.35 m (pitch 34)
Same shot, no tether	4.95 m
Flight time, apex (0.75 N)	0.79 s, 1.10 m
Impact speed, angle (0.75 N)	5.7 m/s, 56 deg down
Ballistics table	6 to 47 deg, best 33.5 deg: 3.71 m

Storage reel payout (LIGHT 0.5 N at full reel)

Reel empty mass, inertia full	46 g, 31.4 kg mm2
Line radius full / core	30.7 / 25.0 mm (brake +23 % at core)
Line pull in the stroke (spin-up)	5.00 N
Reel rpm max	2115 rpm
Overrun after landing	0.56 m in 0.26 s

Capstan and tow

Wraps for tow limit at the capstan (75 N, mu 0.15)	2.9 raw, 5 design
Take-up needed for the tow at the actual wraps (7)	0.10 N
Tow at limit: torque, speed, current	4.17 N m, 0.128 m/s, 6.09 A
Line pull at limiter (5 N m) (12.6 V)	61 N (break 667 N)
Line pull, motor stall (no limiter, 12.6 V)	174 N
Tow sensing at the dart: N per wiper amp (dart reel)	11.3 N/A (dart = 11.3 x (I - I_free) - 2.0 N)
Tow sensing at the capstan (x1.5)	17.0 N/A (capstan = 17.0 x (I - I_free) - 3.0 N)

Drivetrain (wiper at 11.1 V with sag)

aim: peak torque, current, time	0.33 N m, 1.9 A, 7 s
cock: peak torque, current, time	2.93 N m, 4.7 A, 3 s
reel: peak torque, current, time	0.43 N m, 2.0 A, 107 s
tow: peak torque, current, time	4.17 N m, 6.1 A, 39 s
Wiper at limiter (5 N m), 12.6 V (with sag)	5.0 N m, 7.0 A
Wiper motor stall (no limiter), 12.6 V (with sag)	14.1 N m, 17.0 A

Harry must measure (then edit the parameter block and rerun: python summary.py)

- Spring hang test: hang 1, 2, 3 kg from one spring, measure the length each time. Slope = rate, intercept = initial tension. Every range number scales with the stored energy.
- Wiper: free speed (turns in 30 s at 12.0 V), free current, stall torque (arm + luggage scale, 1 s) and stall current. Edit common.Wiper or spec WIPER.
- Dart mass on a kitchen scale with ballast and tether knot (spec 0.279 kg).
- Slip clutch: luggage scale on the line with the reel full: LIGHT 0.5 N, FIRM 5 N.
- After the first shots: python calib.py range pitch:brake:range ... then python ballistics.py --v-scale <factor>.
- Rail friction: tilt the rail with the dart on it until it slides (mu = tan angle).